

Identifying fossil pollen at scale

Small fossils.
A large scientific record.

Smithsonian × Rice D2K

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Pollen reveals ancient vegetation

**~50 million
years ago**

Identify past plant communities.
Support research on warm climates.



Finding a specimen is not the same as naming it

DETECTION

Where is the specimen?

CLASSIFICATION

What category is it?

We will compare two ways of connecting these tasks.

83 slides. 2,290 target annotations.

83

annotated slides

2,290

target annotations

32

candidate categories

Large images with multiple focal depths

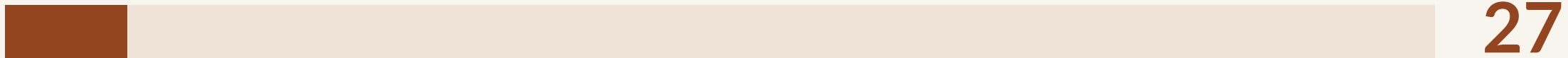
Images stay on Rice's cluster; we read small regions.

Some categories have far fewer examples

Bisaccate



Cupuliferoipollenites sp.



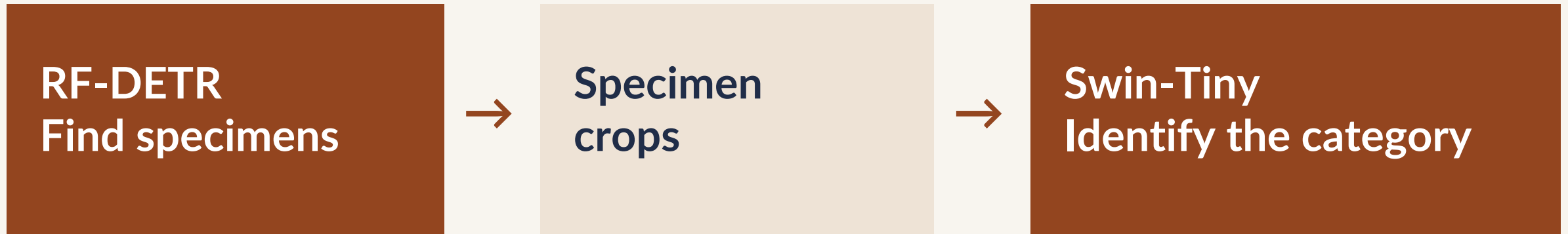
Evaluate each category—not just overall accuracy.

One model finds and identifies



A unified model learns both tasks together.

Find first. Classify the crop.



Separates identification quality from detection quality.

The existing detector can recover known targets

54 / 64

Best focal plane

51 / 64

Focus stack

YOLO checkpoints • 64 targets across 29 slides

A feasibility check—not independent classification performance.

A fair comparison needs unseen source groups

- 01 Separate slides** Keep each specimen and all its views together.
- 02 Use one protocol** Same categories, preprocessing, and evaluation data.
- 03 Count all errors** Measure missed specimens and wrong categories.

From annotated specimens to useful predictions

01

**Inspect crops
and define groups**

02

**Freeze the split
and compare
models**

03

**Build a simple
review interface**

Deliver images, proposed categories, and uncertainty for expert review.